Class Notes/B.Tech/Semester-I/2024-2025

Linear Algebra & Differential Equations (LADE)

UNIT-I: Linear Systems and Matrix Operations

1. Introduction

Reasercher in many other fields have employed computer to analyze mathematical models. Because of the massive amount of data involved the models are usually linear that is, they are described by the system of linear equation. The importance of linear algebra for applications has risen indirect proportion to the increase in computing power with each new generation of hardware and software triggering a demand for even greater capabilities. Computer Science is thus intricately linked with linear algebra through the explosive growth of parallel processing and large scale computations. Scientists and engineers now work on problems for more complex than even dreamed possible a few decade age. Today, linear algebra has more potential value for students in many scientific and business fields than any other undergraduate mathematical subject.

System of linear equations can be used to solve resource allocation problems in business and economics. Such system can involve many equation in many variables. So after reviewing methods for solving two linear equations in two variables, we use matrices and matrix operations to develop procedure that are suitable for solving linear systems of any size.

2. System of linear equations

A linear equation in the variables x_1, x_2,....., x_n is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + ... + a_nx_n = b \tag{2.1}$$

where, b and the coefficients a_1, a_2,....., a_n are real numbers.

Note:

1. A system of linear equations (or a linear system) is a collection of one or more linear equations involving the same set of variables, say x_1,...., x_n, respectively.
2. The set of all possible solutions is called the solution set of linear system.
3. Two linear systems are called equivalent if they have the samr solution set.
4. A system of linear equations has either (i) no solution, (ii) exactly one solution, (iii) infinitely many solutions.
5. A linear system is consistent if it has either one solution or infinitely many solutions.

   A system is inconsistent if it has no solution.

Definition 2.1. ***Linaer System:***
A linear system of 'm' equations in 'n' unknown $x_1, x_2, \ldots x_n$ is a set of equation of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2n}x_n &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \ldots + a_{3n}x_n &= b_3 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \ldots + a_{mn}x_n &= b_m \end{aligned}$$

Note: Linear system is called Homogeneous if $b_1 = 0 = b_2 = \ldots = b_m$ and Non-Homogeneous otherwise.
We rewrite the above equations in the form $Ax = B$, where

$$A(CoefficientMatrix) = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

$$X(VariableMatrix) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

and

$$B(ConstantMatrix) = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

Augmented matrix:

An Augmented matrix for a system of equations is a matrix of numbers in which each row represents the constants from one equation(both the coefficients and constant on the other side of the equal sign) and each column represents all the coefficient for a single variable.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

Example 2.1. *Let the system of linear equations is*

$$x - 2y + 3z = 7 \tag{2.2}$$

$$2x + y + z = 4 \tag{2.3}$$

$$-3x + 2y - 2z = -10 \tag{2.4}$$

The augmented matrix of the above linear system is

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 7 \\ 2 & 1 & 1 & 4 \\ -3 & 2 & -2 & -10 \end{array}\right]$$

2.1. **Row Operations and Equivalent Systems:**

Definition 2.2. ***Elementary Operations*** *The following operations* 1, 2 *and* 3 *are called elementary operations:*

(1) *Interchange of any two equations.*
(2) *Multiply a nonzero constant throught an equation.*
(3) *Replace an equation by itself plus a constant multiple of another equation.*

Definition 2.3. ***Equivalent Linear Systems:*** *Two linear systems are said to be equivalent if one can be obtained from the other by a finite number of elementary operations.*

Definition 2.4. ***Elementary Row Operations:*** *The elementary row operation are defined as:*

1. *Interchange of any two rows, say, "interchange the ith and j^{th} rows", denoted by $R_i \leftrightarrow R_j$.*
2. *Multiply a non zero constant throught a row, say, "multiply the ith row by $C \neq 0$", denoted by $R_i \to CR_i$.*
3. *Replace row by itself plus a constant multiple of another row, say, "replace the ith row by i^{th} row plus C times the jth row", denoted by $R_i \to R_i + CR_j$.*

Definition 2.5. ***Row Equivalent Matrices:*** *Two matrices are said to be row-equivalent if one can be obtained from the other by a finite number of elementary row operations.*

Example 2.2.

$$\begin{bmatrix} 0 & 1 & 1 & 2 \\ 2 & 0 & 3 & 5 \\ 1 & 1 & 1 & 3 \end{bmatrix}$$

$R_1 \leftrightarrow R_2$

$$\sim \begin{bmatrix} 2 & 0 & 3 & 5 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \end{bmatrix}$$

$R_1 \to \frac{1}{2}R_1$

$$\sim \begin{bmatrix} 1 & 0 & \frac{3}{2} & \frac{5}{2} \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \end{bmatrix}$$

$R_3 \to R_3 - R_1$

$$\sim \begin{bmatrix} 1 & 0 & \frac{3}{2} & \frac{5}{2} \\ 0 & 1 & 1 & 2 \\ 0 & -1 & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

All the above four matrices are equivalent matrices.

(1) Use row reduction to solve the following system of linear equations: $x+y+z = 5$;
$x + 3y + z = 9$;
$4x - y + z = -2$.
Solution: We have

$$x + y + z = 5 \tag{2.5}$$

$$x + 3y + z = 9 \tag{2.6}$$

$$4x - y + z = -2 \tag{2.7}$$

Write down the matrix for the system and reduce it to row reduced echelon form:

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 1 & 3 & 1 & 9 \\ 4 & -1 & 1 & -2 \end{array}\right]$$

$R_3 \to R_3 - 4R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 1 & 3 & 1 & 9 \\ 0 & -5 & -3 & -22 \end{array}\right]$$

$R_2 \to R_2 - R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 2 & 0 & 4 \\ 0 & -5 & -3 & -22 \end{array}\right]$$

$R_2 \to \frac{1}{2}R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & 1 & 0 & 2 \\ 0 & -5 & -3 & -22 \end{array}\right]$$

$R_1 \to R_1 - R_2$, $R_3 \to R_3 + 5R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -3 & -12 \end{array}\right]$$

$R_3 \to -\frac{1}{3}R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

$R_1 \to -R_1 - R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

$\therefore$ The solution is $x = -1$, $y = 2$ and $z = 4$.

3. Row reduction and Echelon form

Definition 3.1. *A rectangular matrix is in echelon form (or row echelon form) if it has the following three properties:*

(1) *All non-zero rows occur above any zero rows.*
(2) *Each leading entry of a row in a column occurs to the right of the leading entry of the row above it.*
(3) *All entries in a column below a leading entry are zeros.*

If a matrix in echelon form satisfies the following additional conditions, then it is in reduced echelon form (or reduced row echelon form).

(4) *The leading entry in each non-zero row is* 1.
(5) *Each leading* 1 *is the only non-zero entry in its column.*

Note 1: Echelon form means that the matrix is of either row echelon form or reduced row echelon form.

Note 2: A **Pivot position** in a matrix A is a location in A that corresponds to a leading 1 in the reduced echelon form of A. A **Pivot column** is a column of A that contains a pivot position.

Example 1: The following are examples of matrices in echelon form:

$$\begin{bmatrix} 4 & 2 & 3 \\ 0 & 0 & 5 \end{bmatrix}, \begin{bmatrix} 3 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix}, \begin{bmatrix} 7 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 4 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Example 2: The following are examples of matrices in reduced echelon form:

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

3.1. **Problems:**

(1) Row reduce the following matrix to echelon form and locate the pivot columns of A:

$$A = \begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$$

Ans:

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 1 & 2 & -3 & -3 \\ 0 & 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(2) Apply elementary row operations to transform the following matrix into echelon form. Find pivot positions and columns of A:

$$A = \begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

Ans:

$$A = \begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

(3) Row reduce the following matrix to reduced echelon form. Circle the pivot position in the reduced matrix and list the pivot columns:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix}$$

Ans:

$$A \sim \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(4) Row reduce the following matrix to reduced echelon form. Circle the pivot position in the reduced matrix and list the pivot columns:

$$A = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 3 & 5 & 7 & 9 \\ 5 & 7 & 9 & 1 \end{bmatrix}$$

Ans:

$$A \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(5) Row reduce the following matrix to reduced echelon form. Circle the pivot position in the reduced matrix and list the pivot columns:

$$A = \begin{bmatrix} 1 & 2 & -2 & 3 \\ 2 & 5 & -4 & 6 \\ -1 & -3 & 2 & -2 \\ 2 & 4 & -1 & 6 \end{bmatrix}$$

Ans:

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3.2. **Practice Problem:**

(1) Determine which matrices are in reduced echelon forms and which others are only in echelon forms:

$$(i) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}; (ii) \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}; (iii) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}; (iv) \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix};$$

$$(v) \begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}; (vi) \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}; (vii) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}; (viii) \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix};$$

(2) Row reduce the following matrices into echelon forms:

$$(a) \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}; (b) \begin{bmatrix} 2 & 3 & 4 & -1 \\ 5 & 2 & 0 & -1 \\ -4 & 5 & 12 & -1 \end{bmatrix}; (c) \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$$

(3) Row reduce the following matrices into reduced echelon form:

$$(a) \begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}; (b) \begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}; (c) \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & -4 \\ 2 & 3 & 5 & -5 \\ 3 & -4 & -5 & 8 \end{bmatrix}$$

4. Solution of System of Linear equations

Let the reduced echelon form of a given augmented matrix is obtained as

$$\begin{bmatrix} 1 & 0 & -5 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There are three variables as the augmented matrix has four columns. The associated system of equations can be written from the above matrix as

$$\begin{aligned} x_1 - 5x_3 &= 1 \\ x_2 + x_3 &= 4 \\ 0 &= 0 \end{aligned}$$

The variables x_1 and x_2 corresponding to pivot column in the matrix are called **basic variables**. The other variables, x_3, is called a **free variable.**

Consistency of a system of linear equation:
If a system of linear equations has either one or infinitely many solutions, then it is said to be **consistent**. A system is said to be **inconsistent** if it has no any solution.

Existence and uniqueness theorem for consistency:
A linear system is consistent if an only if an echelon form of the augmented matrix has no row of the form

$$[\, 0 \cdots 0 \; b \,] \text{ where, } b \neq 0.$$

If a linear system is consistent, then the solution set contains either (i) a unique solution, when there are no free variables, or (ii) infinitely many solutions, when there is at least one free variable.

Working Procedure to solve a linear system of equations:

1 Write the Augmented matrix of the system.
2 Find Row echelon form to check the system is consistent. If it inconsistent stop, otherwise next step.
3 Continue to obtain reduced echelon form.
4 Write the system of equations corresponding to the matrix obtained in step (3).
5 Rewrite each non-zero equation from step 4 so that its one basic variable is expressed interms of any free variables appearing in the equation.

5. PROBLEMS:

(1) Determine if the following system is consistent? If so then solve:

$$x_1 - 2x_2 + x_3 = 0$$
$$2x_2 - 8x_3 = 8$$
$$-4x_1 + 5x_2 + 9x_3 = -9$$

Sol: The augmented matrix of the given system is

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{array}\right]$$

$R_4 \to R_4 + 4R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & -3 & 13 & -9 \end{array}\right]$$

$R_2 \to \frac{1}{2}R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{array}\right]$$

$R_3 \to R_3 + 3R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Which is in echelon form and the number of non-zero rows in the coefficient matrix and in the augmented matrix is same (i.e, 3). Therefore, the given system of equations is consistent.

Now, the corresponding system of linear equations of the above echelon form is written as

$$x_1 - 2x_2 + x_3 = 0$$
$$x_2 - 4x_3 = 4$$
$$x_3 = 3$$

The back substitution gives $x_3 = 3, x_2 = 16, x_3 = 29$.

Hence, $(29, 16, 3)$ is the required solution.

Note: *In the above case, the number of non-zero rows of the echelon form is equal*

to the number of unknown variables in the given system of linear equations, therefore there exists unique solution. If the number of non-zero rows of the echelon form is less than the number of unknown variables, then the system will have infinite number of solutions.

(2) *Determine if the following system is consistent? If so then solve:*

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ 5x_1 - 5x_3 &= 10 \end{aligned}$$

Ans: $(1, 0, -1)$.

(3) *Determine if the following system is consistent? If so then solve:*

$$\begin{aligned} x_1 - 4x_2 + 9x_3 &= 1 \\ x_2 + 7x_3 &= 0 \\ 2x_3 &= 0 \end{aligned}$$

Ans: $(1, 0, 0)$.

(4) *Determine if the following system is consistent? If so then solve:*

$$\begin{aligned} x_1 - x_2 &= -4 \\ x_2 - x_3 &= -7 \\ x_3 - 3x_4 &= -1 \\ 2x_4 &= 4 \end{aligned}$$

Ans: $(-6, -2, 5, 2)$.

(5) *Determine if the following system is consistent:*

$$\begin{aligned} x_2 - 4x_3 &= 8 \\ 2x_1 - 3x_2 + 2x_3 &= 1 \\ 5x_1 - 8x_2 + 7x_3 &= 1 \end{aligned}$$

(6) *Determine if the following system is consistent:*

$$x_2 + 4x_3 = -5$$
$$x_1 + 3x_2 + 5x_3 = -2$$
$$3x_1 + 7x_2 + 7x_3 = 6$$

(7) *Determine if the following system is consistent:*

$$x_1 - 3x_2 + 4x_3 = -4$$
$$3x_1 - 7x_2 + 7x_3 = -8$$
$$-4x_1 + 6x_2 - x_3 = 7$$

(8) *Determine if the following system is consistent? If so then solve:*

$$x_1 - 2x_2 - x_3 + 3x_4 = 0$$
$$-2x_1 + 4x_2 + 5x_3 - 5x_4 = 3$$
$$3x_1 - 6x_2 - 6x_3 + 8x_4 = 2$$

(9) *Determine if the following system is consistent? If so then solve:*

$$x_1 + 3x_2 + 4x_3 = 7$$
$$3x_1 + 9x_2 + 7x_3 = 6$$

Infinitely Many Solutions:

(1) Determine if the following system is consistent? If so then determine the general solution of the system:

$$x_1 + 3x_2 + 4x_3 = 7$$
$$3x_1 + 9x_2 + 7x_3 = 6$$

Sol: The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{array}\right]$$

$R_2 \to R_2 - 3R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{array}\right]$$

This is in echelon form. Here, the number of non-zero rows in the coefficient matrix and the augmented matrix is same. Therefore the given system is consistent.

Corresponding to the leading entry 1 and -5, we have x_1 and x_3 are the basic variable.

Whereas, x_2 is the free variable.

Hence, the required general solution can be obtained in term of the free variable. The echelon form can be put back in the system of linear equations as

$$x_1 + 3x_2 + 4x_3 = 7$$
$$-5x_3 = -15$$

Or,

$$x_3 = 3$$
$$x_1 = -5 - 3x_2$$

This is the required general solution, where x_2 is free to choose any value.

(2) Determine the general solution of the system

$$x_1 + 4x_2 + 0x_3 = 7$$
$$2x_1 + 7x_2 + 0x_3 = 10$$

(3) Find the general solution of the system and write in the Parametric form:

$$x_1 - 3x_2 - x_4 = -2$$
$$x_2 - 4x_5 = 1$$
$$x_4 + 9x_5 = 4$$
$$0x_5 = 0$$

(4) Choose h and k such that the system has (a) No Solution; (b) a Unique Solution; (c) Many Solution.

i

$$x_1 + hx_2 = 2$$
$$4x_1 + 8x_2 = k$$

ii

$$x_1 + 3x_2 = 2$$
$$3x_1 + hx_2 = k$$

(5) Find the general solution of the systems whose augmented matrices are given below:

$$(i)\begin{bmatrix} 0 & 1 & -6 & 5 \\ 1 & -2 & 7 & -6 \end{bmatrix}; (ii)\begin{bmatrix} 3 & -4 & 2 & 0 \\ -9 & 12 & -6 & 0 \\ -6 & 8 & -4 & 0 \end{bmatrix};$$

6. Rank of a matrix by Row Echelon form

Definition 6.1. *The rank of a matrix is equal to the number of linearly independent rows or the number of non-zero rows in the echelon form of that matrix. Rank of a matrix A is denoted by $\rho(A)$.*

Note: A linearly independent row is one that is not a combination of other rows.

Working rule to find Rank of a given matrix: Finding the rank of a matrix is simple if you know how to find the row echelon matrix. To find the rank of a matrix:

(1) Find the row echelon matrix in which all the non-zero rows are linearly independent to each other.
(2) Count the number of non-zero rows.
(3) The number of non-zero rows is said to be the rank of the given matrix.

6.1. Problems:

(1) Find the rank of the matrix

$$A = \begin{bmatrix} 5 & 0 & 1 \\ 2 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution: Let

$$A = \begin{bmatrix} 5 & 0 & 1 \\ 2 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$R_2 \to R_2 - \frac{2}{5}R_1$

$$\equiv \begin{bmatrix} 5 & 0 & 1 \\ 0 & 0 & \frac{8}{5} \\ 0 & 0 & 1 \end{bmatrix}$$

$R_3 \to R_3 - \frac{5}{8}R_2$

$$\equiv \begin{bmatrix} 5 & 0 & 1 \\ 0 & 0 & \frac{8}{5} \\ 0 & 0 & 0 \end{bmatrix}$$

The total number of non-zero rows is 2, therefore $\rho(A) = 2$.

(2) Find the rank of the matrix

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$$

Solution: Let

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$$

The order of B is $3X3$, so $\rho(B) \leq 3$.

Let us transform the matrix B to an echelon form by using elementary transformations. $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$

$$\equiv \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix}$$

$R_3 \to R_3 - R_2$

$$\equiv \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

The number of non-zero rows is 2, therefore $\rho(B) = 2$.

(3) Find the rank of the matrix

$$C = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{bmatrix}$$

Ans: $\rho(C) = 3$.

(4) Find the rank of the matrix

$$D = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 3 & 4 & 5 & 2 \\ 2 & 3 & 4 & 0 \end{bmatrix}$$

Ans: $\rho(D) = 3$.

7. Gauss elimination

In this method, the unknowns are eliminated successively and the system is reduced to an upper triangular system from which the unknown are found by back substitution.

7.1. **Working Procedure:**

(1) Write the augmented Matrix.

(2) Reduce the augmented matrix into echelon form.

(3) Put back the obtained echelon form into the new system of equations.

(4) Then by back substitution, we get the values for variables $(x_1, x_2, \cdots x_n)$.

7.2. **Problems:**

(1) Solve $2x - 2y + 3z = 2$; $x + 2y - z = 3$; $3x - y + 2z = 1$ by using Gauss elimination method.

Solution: The given system of equation is

$$2x - 2y + 3z = 2 \tag{7.1}$$

$$x + 2y - z = 3 \tag{7.2}$$

$$3x - y + 2z = 1 \tag{7.3}$$

The augmented matrix of the above system is

$$[A|B] = \left[\begin{array}{ccc|c} 2 & -2 & 3 & 2 \\ 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 1 \end{array}\right]$$

Transfering from augmented matrix to echelon form:

$R_1 \leftrightarrow R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & -2 & 3 & 2 \\ 3 & -1 & 2 & 1 \end{array}\right]$$

$R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -6 & 5 & -4 \\ 0 & -7 & 5 & -8 \end{array}\right]$$

$R_3 \to R_3 - R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -6 & 5 & -4 \\ 0 & -1 & 0 & -4 \end{array}\right]$$

$R_2 \leftrightarrow R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -1 & 0 & -4 \\ 0 & -6 & 5 & -4 \end{array}\right]$$

$R_3 \to R_3 - 6R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -1 & 0 & -4 \\ 0 & 0 & 5 & 20 \end{array}\right]$$

which is in echelon form.

Now, by the row-echelon form the given system of equations reduces to

$$x + 2y - z = 3 \tag{7.4}$$

$$-y = -4 \tag{7.5}$$

$$z = 4 \tag{7.6}$$

On solving the above three equations we have $x = -1$, $y = 4$, $z = 4$.

(2) Solve $2x + 4y + 6z = 22$; $3x + 8y + 5z = 27$; $-x + y + 2z = 2$ by using Gauss elimination method.

Solution: The given system of equation is

$$2x + 4y + 6z = 22 \tag{7.7}$$

$$3x + 8y + 5z = 27 \tag{7.8}$$

$$-x + y + 2z = 2 \tag{7.9}$$

The augmented matrix of the above system is

$$[A|B] = \left[\begin{array}{ccc|c} 2 & 4 & 6 & 22 \\ 3 & 8 & 5 & 27 \\ -1 & 1 & 2 & 2 \end{array}\right]$$

Transfering augmented matrix to echelon form we get $R_1 \to \frac{1}{2}R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 3 & 8 & 5 & 27 \\ -1 & 1 & 2 & 2 \end{array}\right]$$

$R_3 \to R_3 + R_1$, $R_2 \to R_2 - 3R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 2 & -4 & -6 \\ 0 & 3 & 5 & 13 \end{array}\right]$$

$R_2 \to R_2 - 2R_3$,$R_3 \to R_3 - R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 0 & -22 & -44 \\ 0 & 1 & 9 & 19 \end{array}\right]$$

$R_2 \leftrightarrow R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 1 & 9 & 19 \\ 0 & 0 & -22 & -44 \end{array}\right]$$

By row-echelon form the given system of equations reduces to

$$x + 2y + 3z = 11 \tag{7.10}$$

$$y + 9z = 19 \tag{7.11}$$

$$-22z = -44 \tag{7.12}$$

On solving above three equations we have $x = 3$,$y = 1$,$z = 2$.

(3) Solve the linear system by Gauss elimination method

$$y + z = 2$$

$$2x + 3z = 5$$

$$x + y + z = 3$$

7.3. **Practice Problems:**

(1) Apply Gauss elimination method to solve the system of equations: $x_1 - 2x_2 + 3x_3 = 2$; $3x_1 - x_2 + 4x_3 = 4$; $2x_1 + x_2 - 2x_3 = 5$.
Ans: $x_1 = \frac{11}{5}, x_2 = -\frac{7}{5}, x_3 = -1$.

(2) Apply Gauss elimination method to solve the system of equations: $2x_1 + x_2 + 4x_3 = 12$; $4x_1 + 11x_2 - x_3 = 33$; $8x_1 - 3x_2 + 2x_3 = 20$. **Ans:** $x_1 = 3, x_2 = 2, x_3 = 1$.

(3) Apply Gauss elimination method to solve the system of equations: $x_1 + x_2 + x_3 + x_4 = 2$; $2x_1 - x_2 + 2x_3 - x_4 = -5$; $3x_1 + 2x_2 + 3x_3 + 4x_4 = 7$; $x_1 - 2x_2 - 3x_3 + 2x_4 = 5$.
Ans: $x_1 = 0, x_2 = 1, x_3 = -1$.

(4) Apply Gauss elimination method to solve the system of equations: $10w - 7x + 3y + 5z = 6$; $-6w + 8x - y - 4z = 5$; $3w + x + 4y + 11z = 2$; $5w - 9x - 2y + 4z = 7$.
Ans: $w = 5, x = 4, y = -7, z = 1$.

(5) Apply Gauss elimination method to solve the system of equations: $2x_1 + 2x_2 + 4x_3 = 18$; $x_1 + 3x_2 + 2x_3 = 13$; $3x_1 + x_2 + 3x_3 = 14$.

8. Inverse of a matrix by Gauss Jordan Method

The elementary row transformations which reduce a given square matrix A to the unit matrix, when applied to unit matrix I give the inverse of A. **Working rule to evaluate A^{-1}:** Write the two matrices A and I side by side. Then perform the same row transformations on both. As soon as A is reduced to I; the other matrix represents A^{-1}.

8.1. **Problems:**

(1) Using Gauss-Jordan method, find the inverse of the matrix

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$$

Solution: Let

$$A = \left[\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 1 & 3 & -3 & 0 & 1 & 0 \\ -2 & -4 & -4 & 0 & 0 & 1 \end{array}\right]$$

$R_2 \to R_2 - R_1$ and $R_3 \to R_3 + 2R_1$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 0 & 2 & -6 & -1 & 1 & 0 \\ 0 & -2 & 2 & 2 & 0 & 1 \end{array}\right]$$

$R_2 \to \frac{1}{2}R_2$ and $R_3 \to \frac{1}{2}R_3$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -1 & 1 & 1 & 0 & \frac{1}{2} \end{array}\right]$$

$R_1 \to R_1 - R_2$ and $R_3 \to R_3 + R_2$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 6 & \frac{3}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & -3 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -2 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right]$$

$R_1 \to R_1 + 3R_3$,$R_2 \to R_2 - \frac{3}{2}R_3$ and $R_3 \to -\frac{1}{2}R_2$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 1 & \frac{3}{2} \\ 0 & 1 & 0 & -\frac{5}{4} & -\frac{1}{4} & -\frac{3}{4} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{array}\right]$$

Hence the inverse of the given matrix is

$$\begin{bmatrix} 3 & 1 & \frac{3}{2} \\ -\frac{5}{4} & -\frac{1}{4} & -\frac{3}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

(2) Find inverse of the matrix

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Sol:‘Consider the matrix

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$R_1 \to \frac{1}{2}R_1$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

$R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_1$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} & 0 & 1 \end{array}\right]$$

$R_2 \to \frac{2}{3}R_2$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} & 0 & 1 \end{array}\right]$$

$R_3 \to R_3 - \frac{1}{2}R_2$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & \frac{4}{3} & -\frac{1}{3} & -\frac{1}{3} & 1 \end{array}\right]$$

$R_3 \to \frac{3}{4}R_3$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{array}\right]$$

$R_2 \to R_2 - \frac{1}{3}R_3$, $R_1 \to R_1 - \frac{1}{2}R_3$

$$\left[\begin{array}{ccc|ccc} 1 & \frac{1}{2} & 0 & \frac{5}{8} & \frac{1}{8} & -\frac{3}{8} \\ 0 & 1 & 0 & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{array}\right]$$

$R_1 \to R_1 - \frac{1}{2}R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & 1 & 0 & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{array}\right]$$

Hence, the inverse of the given matrix is

$$\left[\begin{array}{ccc} \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{array}\right]$$

.

(3) Find inverse of the matrix using Gauss Jordan method:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 2 \\ 2 & 4 & 7 \end{bmatrix}$$

(4) Find inverse of the matrix using Gauss Jordan method:

$$\begin{bmatrix} 1 & 3 & 3 \\ 2 & 3 & 2 \\ 2 & 4 & 7 \end{bmatrix}$$

(5) Find inverse of the matrix using Gauss Jordan method:

$$\begin{bmatrix} 2 & -1 & 3 \\ -1 & 3 & -2 \\ 2 & 4 & 1 \end{bmatrix}$$

8.2. **Practice Problems:**

(a) Find the inverse of the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$$

Ans:

$$A^{-1} = \begin{bmatrix} -\frac{9}{2} & 7 & -\frac{3}{2} \\ -2 & 4 & -1 \\ \frac{3}{2} & -2 & \frac{1}{2} \end{bmatrix}$$

(b) Find the inverse of the matrix A, if it exists, where

$$A = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 5 & 6 \\ 5 & -4 & 5 \end{bmatrix}$$

Ans: A^{-1} does not exist.

(c) Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Ans:

$$A^{-1} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

9. LU DECOMPOSITION

LU Decomposition of a matrix is the factorization of a given square matrix into two triangular matrices, one upper triangular matrix and one lower triangular matrix, such that the product of these two matrices gives the original matrix.

A square matrix A can be decomposed into two square matrices L and U such that $A = LU$.

Where U is an upper triangular matrix formed as a result of applying the gauss elimination method on A and L is the lower triangular matrix with diagonal elements being equal to 1.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

We have

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$$

and

$$U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

9.1. Steps for LU Decomposition:

(a) Given a set of linear equations, first convert them into matrix from $AX = B$, where A is the coefficient matrix, X is the variable matrix and B is the matrix of numbers on the right-hand side of the equations.

(b) Now, reduce the coefficient matrix A, i.e., the matrix obtained from the coefficients of variables in all the given equations such that for $'n'$ variables we have an nXn matrix to row echelon form using Gauss Elimination method. The matrix so obtained is U.

(c) To find L, we have two methods, the first one is to assume the remaining elements as some artificial variables, make equations using $A = LU$ and solve them to find those artificial variable, and the other method is that the remaining elements are the **multiplier coefficients** because of which the respective positions became zero in the U matrix.

(d) Now we have A (the n x n coefficient matrix), L (the n x n lower triangular matrix), U (the n x n upper triangular matrix), X (the n x 1 matrix of variables) and B (the n x 1 matrix of numbers on the right-hand side of the equations).

(e) In the given system of equations, $AX = B$, we substitute $A = LU$. Thus we have $LUX = B$. We put $UX = Y$, where Y is a variable matrix and solve $LY = C$ for Y first and then finally solve $UX = Y$ for X, which will give the required solution.

9.2. **Problems:**

(a) Solve the following system of equations using LU decomposition method: $2x + 5y = 21$ and $x + 2y = 8$.

Sol: We have

$$2x + 5y = 21 \tag{9.1}$$

$$x + 2y = 8 \tag{9.2}$$

Matrix form of the above linear equations is

$$\begin{bmatrix} 2 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 21 \\ 8 \end{bmatrix}$$

Let

$$A = \begin{bmatrix} 2 & 5 \\ 1 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, B = \begin{bmatrix} 21 \\ 8 \end{bmatrix}$$

Converting the matrix A into upper triangular matrix using Gauss elimination method as follows:

$$A = \begin{bmatrix} 2 & 5 \\ 1 & 2 \end{bmatrix}$$

$R_2 \to R_2 - \frac{1}{2}R_1$

$$= \begin{bmatrix} 2 & 5 \\ 0 & -\frac{1}{2} \end{bmatrix}$$

$$\therefore U = \begin{bmatrix} 2 & 5 \\ 0 & -\frac{1}{2} \end{bmatrix}$$

L is made up of the multiplier used in the above Gauss elimination method with all diagonal elements 1.

$$\therefore\ L = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$$

$\therefore$ LU Decomposition of A is

$$A = \begin{bmatrix} 2 & 5 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 0 & -\frac{1}{2} \end{bmatrix} = LU$$

Now,

$$\begin{aligned} AX &= B \\ \Rightarrow LUX &= B \end{aligned} \tag{9.3}$$

Let $UX = Y$, where $Y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$.

Then from Eq. (9.3), we can write

$$\begin{aligned} LY &= B \\ \Rightarrow \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} &= \begin{bmatrix} 21 \\ 8 \end{bmatrix} \end{aligned}$$

Putting back above matrix form into system of linear equations, we get

$$\begin{aligned} y_1 &= 21 \\ \frac{1}{2}y_1 + y_2 &= 8 \end{aligned}$$

Using $y_1 = 21$, we get $y_2 = -\frac{5}{2}$. Thus,

Now,

$$\begin{aligned} UX &= Y \\ \Rightarrow \begin{bmatrix} 2 & 5 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 21 \\ -\frac{5}{2} \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \Rightarrow 2x + 5y &= 21 \\ -\frac{1}{2}y &= -\frac{5}{2} \end{aligned}$$

On solving we get, $y = 5$ and $x = -2$.

Thus, the solution by LU decomposition method is obtained as $x = -2$ and $y = 5$.

(b) Solve the following system of equations using LU decomposition method: $2x + 5y = 16$ and $3x + 2y = 11$.

Ans: $x = 3$ and $y = 2$.

(c) Solve the following system of equations using LU decomposition method: $2x + 3y - z = 5$, $3x + 2y + z = 10$ and $x - 5y + 3z = 0$.

Sol:

$$2x + 3y - z = 5 \tag{9.4}$$

$$3x + 2y + z = 10 \tag{9.5}$$

$$x - 5y + 3z = 0 \tag{9.6}$$

Matrix form of the given system of linear equations is

$$\begin{bmatrix} 2 & 3 & -1 \\ 3 & 2 & 1 \\ 1 & -5 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 0 \end{bmatrix}$$

Let

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 3 & 2 & 1 \\ 1 & -5 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 5 \\ 10 \\ 0 \end{bmatrix}$$

Converting the matrix A into upper triangular matrix using Gauss elimination method as follows:

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 3 & 2 & 1 \\ 1 & -5 & 3 \end{bmatrix}$$

$R_2 \to R_2 - \frac{3}{2}R_1$

$$= \begin{bmatrix} 2 & 3 & -1 \\ 0 & -\frac{5}{2} & \frac{5}{2} \\ 0 & -5 & 3 \end{bmatrix}$$

$R_3 \to R_3 - \frac{1}{2}R_1$

$$= \begin{bmatrix} 2 & 3 & -1 \\ 0 & -\frac{5}{2} & \frac{5}{2} \\ 0 & -\frac{13}{2} & \frac{7}{2} \end{bmatrix}$$

$R_3 \to R_3 - \frac{13}{5}R_2$

$$= \begin{bmatrix} 2 & 3 & -1 \\ 0 & -\frac{5}{2} & \frac{5}{2} \\ 0 & 0 & -3 \end{bmatrix}$$

$$\therefore U = \begin{bmatrix} 2 & 3 & -1 \\ 0 & -\frac{5}{2} & \frac{5}{2} \\ 0 & 0 & -3 \end{bmatrix}$$

L is made up of the multipliers used in the above Gauss elimination with all diagonal elements 1.

$$\therefore L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ \frac{1}{2} & \frac{13}{5} & 1 \end{bmatrix}$$

$\therefore$ LU Decomposition of A is

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 3 & 2 & 1 \\ 1 & -5 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ \frac{1}{2} & \frac{13}{5} & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & -1 \\ 0 & -\frac{5}{2} & \frac{5}{2} \\ 0 & 0 & -3 \end{bmatrix} = LU$$

Now,

$$AX = B$$
$$\Rightarrow LUX = B$$

Let $UX = Y$, then we can write $LY = B$.

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ \frac{1}{2} & \frac{13}{5} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 0 \end{bmatrix}$$

$$\Rightarrow y_1 = 5$$
$$\frac{3}{2}y_1 + y_2 = 10$$
$$\frac{1}{2}y_1 + \frac{13}{5}y_2 + y_3 = 0$$

and on solving we get, $y_1 = 5$, $y_2 = -\frac{5}{2}$ and $y_3 = -9$.

Now,

$$UX = Y$$

$$\Rightarrow \begin{bmatrix} 2 & 3 & -1 \\ 0 & -\frac{5}{2} & \frac{5}{2} \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ \frac{5}{2} \\ -9 \end{bmatrix}$$

Putting back into system of linear equations, we get

$$\begin{aligned} 2x + 3y - z &= 5 \\ -\frac{5}{2}y + \frac{5}{2}z &= \frac{5}{2} \\ -3z &= -9 \end{aligned}$$

On solving we get; $z = 3$, $y = 2$ and $x = 1$.

Thus, the solution by LU decomposition method is $x = 1$, $y = 2$ and $z = 3$.

(d) Solve the following system of equations using LU decomposition method: $x + y + z = 5$, $2x - y - z = 3$ and $x - y + z = 9$.
Ans: $x = 2$, $y = -3$ and $z = 4$.

(e) Solve the following system of equations using LU decomposition method: $x_1 + x_2 + x_3 = 1$, $4x_1 + 3x_2 - x_3 = 6$ and $3x_1 + 5x_2 + 3x_3 = 4$.
Ans: $x_1 = 1$, $x_2 = 0.5$ and $x_3 = -0.5$.

9.3. **Practice Problems:** Use LU decomposition method to solve the following system of linear equations:

(1)

$$\begin{aligned} 2x + y + 4z &= 12 \\ 4x + 11y - z &= 33 \\ 8x - 3y + 2z &= 20 \end{aligned}$$

Ans: $(x, y, z) = (3, 2, 1)$.

(2)

$$\begin{aligned} 4x + y + z &= 4 \\ x + 4y - 2z &= 4 \\ 3x + 2y - 4z &= 6 \end{aligned}$$

Ans: $(x, y, z) = (1, \frac{1}{2}, -\frac{1}{2})$.

(3)

$$\begin{aligned} x + 5y + z &= 14 \\ 2x + y + 3z &= 13 \\ 3x + y + 4z &= 17 \end{aligned}$$

Ans: $(x, y, z) = (1, 2, 3)$.

(4)

$$\begin{aligned} 3x + 2y + 7z &= 4 \\ 2x + 3y + z &= 5 \\ 3x + 4y + z &= 7 \end{aligned}$$

Ans: $(x, y, z) = (\frac{7}{8}, \frac{9}{8}, -\frac{1}{8})$.

(5)

$$\begin{aligned} 3x + 5y + 2z &= 8 \\ 8y + 2z &= -7 \\ 6x + 2y + 8z &= 26 \end{aligned}$$